



AeroPulse-NG — Field Deployment Kit

Station hardening · Outdoor mast · Industrial compute · Power continuity

AeroPulse-NG v2.4
Prototype Buy Sheet

FIELD DEPLOYMENT KIT

Adds to bench prototype

Outdoor mast · LMR-400 · Filters · Mini-PC · UPS

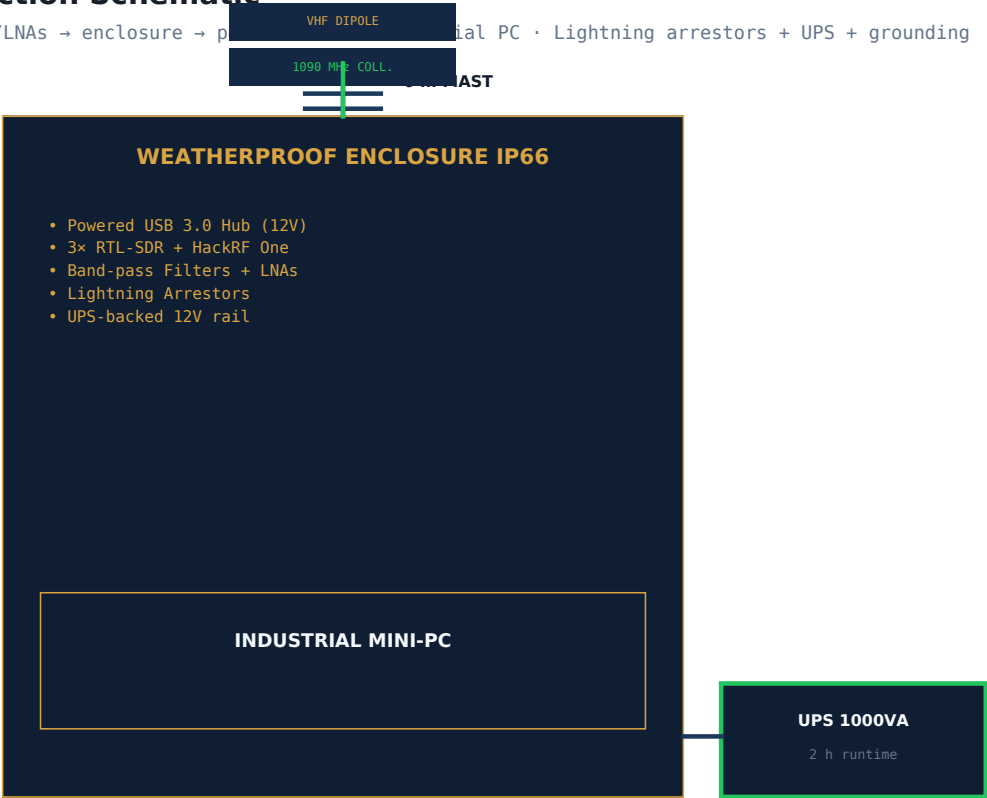
Item	Qty	Unit	Total	Notes
Prototyping kit (above) — reused	1	\$409	\$409	Core is the field core
Industrial Mini-PC i5/16GB/512GB NVMe	1	\$450	\$450	Fanless · on-site edge compute
1090 MHz Collinear 5 dBi fibreglass	1	\$180	\$180	N-female · 1090 MHz ADS-B
VHF Airband Dipole 118-137 MHz	1	\$120	\$120	N-female · VHF comms
LMR-400 Coax 15 m N-male	2	\$85	\$170	Low-loss feedlines
1090 MHz Band-pass Filter + LNA	1	\$95	\$95	Pre-filter + gain
VHF Band-pass Filter + LNA	1	\$85	\$85	Clean VHF RX/TX
Lightning Arrestor N-female	2	\$45	\$90	Surge protection
Mast 6 m telescopic, guyed	1	\$350	\$350	Rooftop / tower mount
Guy wires / clamps / grounding kit	1	\$120	\$120	Installation hardware
UPS 1000 VA (2 h runtime)	1	\$220	\$220	Power continuity
Weatherproof Enclosure IP66	1	\$180	\$180	400×300×150 mm
FIELD SUBTOTAL		\$1,140	Total field kit (prototype + field) \$1,549 ≈ ¥2.4M	

Field kit is deferred until after NDAIE quarter-final. All bench hardware moves to the mast — purchase is not wasted. Outdoor antennas replace indoor whips; industrial PC replaces laptop host.



Field Station — Connection Schematic

Outdoor mast → LMR-400 → filters/LNAs → enclosure → powerline → industrial PC · Lightning arrestors + UPS + grounding



LIGHTNING: Arrestors before enclosure entry · Ground rod <10 Ω · Guy wires bonded.

All bench SDRs move to enclosure; laptop replaced by industrial PC as host; whips replaced by outdoor collinear/dipole.